

INF515: Comparative Genomics

Branch Test for Positive Selection

The objective of this exercise is to use three LRTs to evaluate the following hypotheses: (1) the mutation rate of Ldh-C has increased relative to Ldh-A, (2) a burst of positive selection for functional divergence occurred following the duplication event that gave rise to Ldh-C, and (3) there was a long term shift in selective constraints following the duplication event that gave rise to Ldh-C.

1. Obtain the files for Exercise 3 from the course web-site ([ex3_codeml.ctl](#), [ex3_seqfile.txt](#), [treeH0.txt](#), [treeH1.txt](#), [treeH2.txt](#), [treeH3.txt](#)). The tree files represent different hypotheses denoted H0, H1, H2 & H3 ([LDH_tree.pdf](#)). These hypotheses represent the following concepts:
 1. H0: homogeneous selection pressure over the tree.
 2. H1: episodic change in selection pressure in Ldh-C (only in the branch that immediately follows the gene duplication event).
 3. H2: Long term shift in selection pressure in Ldh-C only; Ldh-C has a permanent change in selection pressure (as compared to its ancestors) whereas Ldh-A remains subject to the ancestral level of selection pressure.
 4. H3: Long term shift in selection on both Ldh-C and Ldh-A; those lineages are subject to selection pressures different from each other and from the ancestor.
2. Run CODEML using the settings in the control file for Exercise 3. Familiarize yourself with the results ([ex3_HelpFile.pdf](#)). In addition to the likelihood score you must be able to identify the branch-specific estimates of the omega parameter. (In the first run, the branch specific values for omega will all be the same. In later runs there will be differences among some branches).
3. You will need to change the control files and re-run CODEML. The objective is to compute the likelihood, and estimate omega parameters, under different models of how selection pressure changes in different parts of the tree. Because the relevant model information is contained in the tree file, you will need several tree files (obtained from the course web site) and change the control file so that it reads the different tree files.

1. As always, you should change the name of the main result file (via **outfile=** in the control file) or you will overwrite your previous results.
2. Change the model assumptions about branch specific omega values by changing the tree files (via **treefile=** and **model=**) set within the control file.
4. Repeat step 3 for each of the four tree files that have been provided to you. Again, keep track of your results by using a table like “Table E3” shown in the slides ([TableE3.pdf](#)), provided below.

Models	ω_{A0}	ω_{A1}	ω_{C1}	ω_{C0}	lnL
H0		NA	NA	NA	
H1		NA	NA		
H2		NA		NA	
H3				NA	

5. In addition, carry out likelihood ratio tests (LRT) of the hypotheses below. Use 1 degree of freedom for each LRT. Helpful: [Chi-Square Calculator](#).
 1. H0 vs. H1:
 2. H0 vs. H2:
 3. H2 vs. H3:

Interpret the results of the LRTs with respect to hypotheses 1 through 3. Which hypothesis is supported by the branch tests? Explain your answer.